

Comparing classical, Bayesian, and machine-learning methods for estimating heterogeneous treatment effects in synthetic obesity trials

by

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Abstract

This study assesses how accurately a classical logistic regression model, a Bayesian hierarchical model (BHM), and a causal forest model estimate heterogeneous treatment effects (HTEs) in synthetic obesity trials inspired by STEP UP. In the simulated trials, participants are randomly assigned to semaglutide or placebo, and the binary outcome is achievement of at least 5% weight loss. Four scenarios represent different forms of HTE: no HTE, a joint subgroup effect, a pattern informed by published subgroup estimates, and continuous age-related heterogeneity. Evaluation uses root mean squared error (RMSE) for individual conditional average treatment effects (CATEs) and prespecified subgroup differences on the risk-difference scale, alongside empirical 95% interval coverage for subgroup differences. The BHM achieves the lowest CATE RMSE in the first three scenarios and the lowest subgroup-difference RMSE under no HTE and joint subgroup heterogeneity. Causal forest achieves the lowest RMSE for both estimation targets in the continuous scenario. In the published scenario, subgroup-difference RMSE is similar across methods, but the classical logistic regression model has coverage closest to the nominal level at 92.8%. Overall, the findings suggest that no model dominates and that model performance depends on both the true form of HTE and the estimation target.

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List of Abbreviations

Abbreviation	Meaning
ATE	average treatment effect
BHM	Bayesian hierarchical model
BMI	body mass index
CATE	conditional average treatment effect
HTE	heterogeneous treatment effect
RCT	randomised controlled trial
RMSE	root mean squared error
STEP	clinical trial programme name
STEP UP	clinical trial name

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Chapter 1

INTRODUCTION

Obesity is a major and growing global public health challenge. Between 1990 and 2021, the prevalence of overweight and obesity among adults increased worldwide (GBD 2021 Adult BMI Collaborators, 2025). Over this period, global deaths and disability-adjusted life years attributable to high body mass index (BMI) have grown significantly (Zhou et al., 2024). Pharmacological options for weight management continue to expand (Wharton et al., 2025), and the STEP 1 trial finds substantial average reductions in bodyweight with once-weekly 2.4 mg semaglutide plus lifestyle intervention compared to placebo (Wilding et al., 2021). The STEP UP trial provides the clinical setting for this study, where researchers test the efficacy and safety of a higher dose of semaglutide (7.2 mg) in adults with obesity but without diabetes. They find that the higher dose of 7.2 mg is more effective in weight reduction than the lower dose of 2.4 mg and placebo in adults with obesity (Wharton et al., 2025).

Randomised trials typically estimate average treatment effects (ATEs) within their study populations, but this may not describe the treatment effect across every patient profile. Therefore, relying only on ATEs may mask important differences between groups of patients, as some groups may benefit more or less from treatment than the average effect suggests (Feuerriegel et al., 2024).

Participants differ in their baseline characteristics, and treatment effects may therefore vary across patient profiles. The question is whether the outcome difference between the treatment and control groups varies systematically with baseline patient characteristics. Systematic variation of this kind is referred to as heterogeneous treatment effects (HTEs) (Rekkas et al.,

2020).

Individual treatment effects cannot be observed directly because, for each participant, only the outcome under the treatment actually received is observed, whilst the potential outcome under the alternative treatment remains unseen (Lamont et al., 2018). The conditional average treatment effect (CATE) is the average treatment effect for individuals with the same baseline characteristics. It is not an observed response (Rekkas et al., 2020). In this study, the CATE is the difference between conditional outcome probabilities on the risk-difference scale. However, sparse subgroups, numerous interactions associated with the curse of dimensionality, and strongly unequal treatment allocation can make estimation less accurate and more unstable (Rekkas et al., 2020; Inoue et al., 2024; Wang et al., 2024). In addition, performing multiple traditional subgroup analyses can introduce the issue of multiplicity, as a larger number of tests increases the probability of false-positive findings (Inoue et al., 2024; Selby et al., 2025).

Parametric models such as classical logistic regression can perform well when their assumed interaction form is correct, but they are less flexible, and any omitted terms can introduce bias (Feuerriegel et al., 2024; Lipkovich et al., 2024). Bayesian hierarchical models (BHMs) can account for subgroup-specific variation through partial pooling (Wang et al., 2024), whilst causal forests can learn nonlinearities and complex HTE patterns without specifying their exact form in advance (Wager and Athey, 2018). However, greater flexibility may increase variance and the risk of overfitting when the sample size is small (Feuerriegel et al., 2024; Van Vogt et al., 2025).

Three methods are used in this study for comparison. The classical logistic regression model contains subgroup indicators derived from the baseline covariates and their interactions with the treatment term. This model relies on correctly specifying the relevant interactions between treatment and the covariates, which is difficult in many settings (Feuerriegel et al., 2024). The BHM estimates outcome risks separately in each treatment arm and assumes that the covariate-defined strata are exchangeable within each arm. This means that the stratum-specific deviations are treated as draws from a common distribution rather than as having distinct prior distributions (Wang et al., 2024). This assumption allows partial pooling, which shrinks stratum-specific log odds towards the arm-specific overall mean log odds when stratum sizes are small and between-stratum variability is limited. This may reduce variance in sparse strata, although shrinkage can introduce bias when true stratum-specific response probabilities differ substantially (Wang et al., 2024). The causal forest model allows nonlinearities and complex patterns

to be learned without specifying their exact form in advance and is therefore less dependent on a prespecified parametric form (Feuerriegel et al., 2024). However, this model generally requires a much larger sample size (Feuerriegel et al., 2024). In this study, all three models are evaluated according to their performance in estimating individual CATEs and subgroup differences under different scenarios generated by simulation.

Existing literature uses different methods to study HTE. Sun et al. (2024) compare classical regression, penalised regression, and tree-based models for estimating HTE. Van Vogt et al. (2025) compare logistic regression, hierarchical lasso logistic regression, and causal forest for identifying HTE by serum potassium. However, a gap remains in the direct comparison of a classical logistic regression model, a BHM, and a causal forest in an unbalanced binary-outcome obesity randomised controlled trial (RCT) setting. This study compares these methods under four simulated scenarios: no HTE, a joint subgroup effect, an HTE pattern constructed from the STEP UP trial summary statistics, and continuous age-related heterogeneity. Model performance is evaluated mainly by comparing predicted individual CATEs and prespecified subgroup differences to their known true values.

This leads to the following research question: How accurately do classical, Bayesian, and machine-learning methods estimate heterogeneous treatment effects under different patterns of heterogeneity in synthetic obesity trials?

The remaining chapters are organised as follows. Chapter 2 reviews the relevant literature. Chapter 3 describes the STEP UP trial and the construction of the synthetic data. Chapter 4 presents the simulation design, estimation methods, and evaluation methods. Chapter 5 reports the results, and Chapter 6 discusses the study’s limitations and implications for method choice in similar settings.

Chapter 2

LITERATURE REVIEW

A central distinction in HTE analysis is between prognostic and predictive factors. Lipkovich et al. (2024) describe prognostic factors as variables associated with outcomes under no treatment, whereas predictive factors are associated with variation in treatment effects. A poor prognosis does not therefore imply greater treatment benefit. The distinction is further complicated by the choice of effect scale. For binary outcomes, variation in baseline risk can produce different absolute effects even when the relative treatment effect is constant (Rekkas et al., 2020). HTE should therefore be interpreted only after the effect scale has been specified.

The unobserved counterfactual outcome also limits direct evaluation of HTE estimators. Simulation addresses this problem by fixing the data-generating process, so the true conditional effects are known. Lamont et al. (2018) use this approach to assess predicted individual treatment effects by comparing estimates with their known values. Sparse subgroups can produce highly variable estimates (Wang et al., 2024), and unequal arm sizes can destabilise some estimators (Inoue et al., 2024). Both features are relevant to the simulated trial design considered here.

Among the methods compared in this study, the classical logistic regression model imposes the strongest prespecified structure. Its estimates can be efficient when the parametric form is correct, but this advantage depends on specifying the relevant interactions and nonlinearities in advance. If the model omits a threshold, joint subgroup relationship, or other effect-modification pattern, it may fail to recover the true CATE pattern (Feuerriegel et al., 2024; Lipkovich et al., 2024). The model therefore offers a clear and interpretable benchmark, but its performance may deteriorate when the

true effect structure does not match the fitted specification. BHMs address sparse-data instability through partial pooling. This relies on an exchangeability assumption, under which the stratum-specific deviations are treated as draws from a common distribution rather than as having distinct prior distributions. Information can therefore be shared across strata, which can reduce variance when this assumption is reasonable (Wang et al., 2024). The same shrinkage can, however, attenuate large genuine differences and produce estimates that are precise but biased when strata are not sufficiently similar (Wang et al., 2024). In the implementation used here, the BHM pools arm-specific outcome probabilities across covariate-defined strata and then derives CATEs from their differences. This is distinct from directly pooling subgroup treatment effects. The prior distributions determine how strongly the strata are pooled (Wang et al., 2024), which motivates the prior sensitivity analysis.

Causal forests approach the problem with less prespecified structure. They use repeated, data-driven partitioning to estimate local treatment effects without requiring a prespecified parametric form (Wager and Athey, 2018). This flexibility can help when HTE depends on complex interactions or nonlinear covariate relationships (Feuerriegel et al., 2024). However, flexible estimation generally requires more data. Small samples and weak treatment-effect signals can therefore increase variability and reduce the accuracy of CATE estimates (Sun et al., 2024; Van Vogt et al., 2025).

Comparative studies support the view that method performance depends on both the data-generating process and the task. Sun et al. (2024) compare regression, penalised regression, and tree-based methods, and show that their performance differs across HTE detection, effect-modifier identification, and subgroup estimation. Lipkovich et al. (2024) likewise find that no one method dominates under all simulation settings. In a semi-synthetic observational study, Wendling et al. (2018) find that causal forests are less competitive when CATE variation is low but perform more strongly under complex HTE. Van Vogt et al. (2025) find that classical regression models, hierarchical lasso, and causal forests all identify potassium as an effect modifier in the empirical application, but only the proposed root-split causal-forest method recovers an unknown threshold in the corresponding simulation. Flexible methods may therefore reveal structures that a prespecified model misses, although this advantage is not uniform across settings.

The literature suggests that no single method performs best in every HTE setting (Lipkovich et al., 2024; Sun et al., 2024). Classical regression may perform well when its interaction structure matches the true treatment-

effect pattern (Feuerriegel et al., 2024; Lipkovich et al., 2024). BHMs may improve stability in sparse subgroups or strata, but their performance depends on whether pooling is appropriate (Wang et al., 2024). Causal forests may better capture complex HTE patterns, but they generally require more data and may have higher variance (Feuerriegel et al., 2024; Van Vogt et al., 2025). These trade-offs motivate the four simulation scenarios used in this study: no HTE, a joint subgroup effect, a pattern informed by STEP UP subgroup estimates, and continuous age-related effect modification. Comparing the methods across these different settings helps assess how their performance depends on the underlying HTE pattern, rather than treating estimation accuracy in one setting as evidence of general superiority.

Within the obesity literature, evidence from semaglutide trials suggests that treatment effects may vary across patient characteristics. In an analysis of the SELECT trial, Ryan et al. (2024) report differences in weight-loss effects across several baseline characteristics, including sex, age, BMI, race, and region, although some of these differences are small. The STEP UP trial provides further subgroup evidence of possible treatment-effect variation for the higher 7.2 mg dose. Published subgroup estimates vary across baseline characteristics, but the reported analyses do not provide clear evidence of HTE (Wharton et al., 2025). Nonetheless, the published STEP UP subgroup summary statistics provide a realistic pattern of heterogeneity that can be used to construct a simulation in which the true conditional treatment effects are known.

Overall, the reviewed literature shows that model performance depends on the underlying HTE structure, the amount of available data, and the evaluation target. Existing comparative studies examine overlapping sets of methods, but among the studies reviewed, no study directly compares classical logistic regression, arm-specific Bayesian partial pooling, and causal forest within the same binary-outcome RCT setting with sparse subgroups under different simulated scenarios. This leaves a gap in understanding how these methods compare when both individual CATEs and prespecified subgroup differences are evaluated against known true values in a realistic obesity setting.

Chapter 3

DATA GENERATION

3.1 The STEP UP trial

The data used in this study are fully synthetic and are informed by the published summary statistics of the STEP UP trial paper (Wharton et al., 2025). No individual-level data from the actual trial are used. The simulation retains selected features of the trial population and treatment comparison, and the baseline distributions and outcome-generating process are specified within this study.

Wharton et al. (2025) describe STEP UP as a randomised, double-blind, phase 3b trial in adults with obesity but without diabetes. Participants are eligible from age 18 years and require a BMI of at least 30 kg/m². The trial randomly assigns 1,407 participants to once-weekly subcutaneous semaglutide 7.2 mg, semaglutide 2.4 mg, or placebo in a 5 : 1 : 1 ratio. All groups receive a lifestyle intervention. Achievement of at least 5% weight loss from baseline to week 72 is one of the coprimary endpoints for the comparison of semaglutide 7.2 mg with placebo.

In this study, the synthetic dataset uses the semaglutide 7.2 mg and placebo arms from the STEP UP trial, with sample sizes of 1,005 and 201, respectively. Each simulated trial therefore contains 1,206 participants with a 5 : 1 treatment allocation. The semaglutide 2.4 mg arm is excluded. In the remaining chapters, semaglutide refers to the 7.2 mg dose unless otherwise stated.

3.2 Synthetic baseline covariates

Four baseline covariates are generated for each participant: age, sex, BMI, and prediabetes status. Age and BMI are continuous, and sex and prediabetes are binary. The published summary statistics of the semaglutide 7.2 mg and placebo arms provide the information used to construct these variables.

The summary statistics table in Wharton et al. (2025) reports 753 female participants in the semaglutide arm and 147 in the placebo arm. Pooling these counts gives 900 female participants among the 1,206 participants in the two arms. Therefore, the synthetic female indicator is generated from a Bernoulli distribution with probability

$$p_{\text{female}} = \frac{753+147}{1206} = \frac{900}{1206},$$

where one denotes female and zero denotes male. This probability determines the expected proportion of female participants, but their number varies between simulated trials.

The corresponding prediabetes counts are 378 in the semaglutide arm and 82 in the placebo arm. The published normoglycaemia counts are 627 and 118, respectively. These categories account for all 1,005 semaglutide participants but only 200 of the 201 placebo participants. The pooled denominator for glycaemic status is therefore 1,205. Prediabetes is generated from a Bernoulli distribution with probability

$$p_{\text{prediabetes}} = \frac{378+82}{1205} = \frac{460}{1205},$$

where one denotes prediabetes and zero denotes normoglycaemia.

For age, the published mean and standard deviation are 47 and 12 years in the semaglutide arm, and 48 and 12 years in the placebo arm. Since the overall mean and standard deviation of the two arms are not published, this study pools the arm-specific statistics to obtain a target mean of approximately 47.17 years and a standard deviation of approximately 12 years. Age is generated from a normal distribution truncated below 18 years. After calibration, the pre-truncation mean and standard deviation are approximately 46.83 and 12.40 years, respectively, so that the resulting truncated distribution matches the pooled age distribution. These pre-truncation values are simulation parameters rather than published STEP UP statistics.

Similarly, for BMI, the published mean and standard deviation are 39.8 and 7.0 kg/m² in the semaglutide arm, and 39.7 and 6.6 kg/m² in the placebo arm. Pooling them gives a target mean of approximately 39.78 kg/m² and a

standard deviation of approximately 6.9 kg/m². BMI is generated from a normal distribution truncated below 30 kg/m². The calibrated pre-truncation mean and standard deviation are approximately 33.81 and 10.32 kg/m², respectively, so that the resulting truncated distribution matches the pooled BMI distribution.

All four covariates are generated independently. In particular, prediabetes probability does not depend on a participant’s generated age, sex, or BMI. Independence and the truncated normal distributions are modelling choices made in this study. They produce the synthetic dataset using information from the STEP UP trial without modelling the joint distribution of the covariates.

Subgroup categories are derived from the four baseline covariates. Age is classified as under 65 years or 65 years or older. BMI is classified as 30 to under 35, 35 to under 40, or 40 or higher, in kg/m². Sex and glycaemic status are classified using their corresponding binary indicators. These labels are all generated from their corresponding baseline covariates.

3.3 Synthetic dataset generation

Each Monte Carlo replication begins with a new baseline sample of 1,206 synthetic participants. Exactly 1,005 participants are randomly assigned to semaglutide and 201 to placebo, independently of their baseline characteristics.

The observed outcome is a binary indicator of achieving at least 5% weight loss. Wharton et al. (2025) report that 862 participants in the semaglutide arm and 63 in the placebo arm achieve this. Therefore, this study defines the response targets as

$$r_{\text{semaglutide}} = \frac{862}{1005} \approx 0.8577, \quad r_{\text{placebo}} = \frac{63}{201} \approx 0.3134.$$

Their difference,

$$\Delta_{\text{target}} = \frac{862}{1005} - \frac{63}{201} \approx 0.5443,$$

defines the target ATE for the synthetic population. These are calibration targets based on the published STEP UP data. The calibration process

is described in Chapter 4.

For each participant, the data-generating process specifies the probability of achieving at least 5% weight loss under semaglutide and under placebo, given the participant’s baseline covariates. The probability corresponding to the treatment actually assigned is then used to generate the observed binary outcome. The four baseline covariates, treatment assignment, and observed outcome are used for model fitting. The underlying treatment-specific response probabilities are retained separately to define the true CATEs used for evaluation.

All participants have complete baseline covariate and outcome data, with no separate generation of missingness, treatment discontinuation, or changes in treatment exposure. Within each scenario and replication, all three estimation methods are fitted to the same realised trial dataset. A new baseline sample, treatment assignment, and outcome are generated for the next replication.

Synthetic datasets are generated under four prespecified scenarios: no HTE, joint subgroup, published scenario, and continuous. Chapter 4 describes their construction, together with the estimation methods and model evaluation.

Chapter 4

METHODS

4.1 Simulation study design

This study compares a classical logistic regression model, BHM, and causal forest under four scenarios of HTE: no HTE, joint subgroup, published, and continuous scenarios. Each scenario contains 500 Monte Carlo replications. Within each replication, a new synthetic trial of 1,206 participants is generated using the process in Chapter 3. All three methods are fitted to the same realised trial dataset, giving 6,000 model fits in the main simulation.

The data-generating process specifies both treatment and control response probabilities, so the true CATE is known for every participant. Each fitted model estimates CATEs in the same fixed evaluation population of 1,000 participants generated independently of the trial datasets. This avoids evaluating the models on the same data used for fitting and assesses how well their estimated CATEs generalise to new participants.

4.2 Placebo model and calibration

Let X_i contain participant i 's baseline covariates and let $Y_i(a)$ denote the potential binary response under arm a , where $a = 0$ denotes placebo and $a = 1$ denotes semaglutide. A response is achievement of at least 5% weight loss. Define

$$p_a(x) = \mathbb{P}(Y_i(a) = 1 \mid X_i = x), \quad \tau(x) = p_1(x) - p_0(x).$$

Thus, $\tau(X_i)$ is the CATE on the risk-difference scale.

The placebo model uses five indicators derived from the four baseline covariates,

$$\begin{aligned} F_i &= \mathbb{I}(\text{female}), & A_i &= \mathbb{I}(\text{age} \geq 65), \\ B_{35,i} &= \mathbb{I}(35 \leq \text{BMI} < 40), & B_{40,i} &= \mathbb{I}(\text{BMI} \geq 40), \\ P_i &= \mathbb{I}(\text{prediabetes}). \end{aligned}$$

Age is measured in years and BMI in kg/m^2 . The reference categories are male, age under 65 years, BMI 30 to under 35, and normoglycaemia. Write $S_i = (F_i, A_i, B_{35,i}, B_{40,i}, P_i)^\top$. The common placebo component is

$$m(X_i) = \lambda_F F_i + \lambda_A A_i + \lambda_{35} B_{35,i} + \lambda_{40} B_{40,i} + \lambda_P P_i.$$

The coefficients are computed from placebo subgroup odds ratios and multiplied by 0.6. For subgroup j with outcome count r_j , subgroup total n_j , and corresponding reference group values $r_{\text{ref},j}$ and $n_{\text{ref},j}$,

$$\lambda_j = 0.6 \log \left(\frac{r_j / (n_j - r_j)}{r_{\text{ref},j} / (n_{\text{ref},j} - r_{\text{ref},j})} \right).$$

For sex, the inputs are 50 women achieving at least 5% weight loss among 147 women and 13 men achieving this among 54 men. For glycaemic status, they are 30 among 82 participants with prediabetes and 33 among 118 with normoglycaemia.

The age and BMI subgroup totals are derived from a calibration population of $N_{\text{cal}} = 200,000$ synthetic participants. For any subgroup g , let $\hat{q}_g$ denote the proportion of the N_{cal} participants belonging to subgroup g , and $\tilde{n}_g$ the corresponding estimated placebo subgroup total. Since the placebo arm contains 201 participants,

$$\hat{q}_g = \frac{1}{N_{\text{cal}}} \sum_{i=1}^{N_{\text{cal}}} \mathbb{I}(X_i \in g), \quad \tilde{n}_g = 201 \hat{q}_g.$$

The age outcome counts are 6 for age 65 or older and 57 for age under 65. The BMI outcome counts are 16, 19, and 28 for BMI 30 to under 35, 35 to under 40, and 40 or higher, respectively. These values define the coefficients used to generate the outcomes.

The placebo subgroup coefficients are scaled by 0.6 to address potentially high placebo probabilities. This allows the constant risk difference in

the no-HTE scenario to be added without producing semaglutide probabilities above 1. The placebo response probability is

$$p_0(X_i) = \text{logit}^{-1}(\alpha_0 + m(X_i)), \quad \text{logit}^{-1}(u) = \frac{1}{1 + \exp(-u)}.$$

Let $r_0 = r_{\text{placebo}}$ and $r_1 = r_{\text{semaglutide}}$. The response targets from Chapter 3 are therefore $r_0 = 63/201$ and $r_1 = 862/1005$. These are simulation calibration targets derived from the STEP UP trial.

The intercept α_0 solves

$$\frac{1}{N_{\text{cal}}} \sum_{i=1}^{N_{\text{cal}}} \text{logit}^{-1}(\alpha_0 + m(X_i)) = r_0.$$

The calibration process uses 80 bisection iterations over the initial interval $[-10, 10]$ and takes the final midpoint as α_0 . The calibration population is generated once and remains fixed throughout the simulation, separate from both the simulated trials and the evaluation population. The semaglutide model differs across the four scenarios, as described next.

4.3 Heterogeneity scenarios

No HTE. Let $\Delta = \Delta_{\text{target}} = r_1 - r_0 = 862/1005 - 63/201$. The semaglutide probability is computed by

$$p_1(X_i) = p_0(X_i) + \Delta, \quad \tau(X_i) = \Delta.$$

The treatment effect is therefore constant on the risk-difference scale. No further calibration is required because, with $\Delta = r_1 - r_0$, calibrating the mean placebo response probability to r_0 also makes the mean semaglutide response probability equal to r_1 in the calibration population. Semaglutide probabilities are checked to ensure that they lie in $[0, 1]$.

Joint subgroup. Define

$$J_i = \mathbb{I}(\text{BMI}_i \geq 40 \text{ and } P_i = 1).$$

The semaglutide model is

$$p_1(X_i) = \text{logit}^{-1}(\alpha_0 + m(X_i) + \alpha_J + 2J_i).$$

Here, J_i acts as a prespecified treatment-effect modifier, with coefficient 2 increasing the semaglutide treatment effect on the log-odds scale for participants with both BMI of at least 40 and prediabetes. This assumes that these participants have a greater response to semaglutide. The calibration intercept α_J is added to the placebo linear predictor $\alpha_0 + m(X_i)$ and is chosen using the same calibration population and bisection method so that the mean semaglutide response probability equals r_1 . Although J_i is binary, participants within the same joint subgroup category can still have different CATEs because their placebo risks may differ.

Published scenario. The published subgroup treatment odds ratios are 11.4 for women and 14.4 for men, 12.9 for age under 65 and 4.6 for age 65 or older, 11.4, 15.7, and 10.7 for the three BMI categories, and 14.7 for normoglycaemia and 9.1 for prediabetes. For each non-reference category, the interaction coefficient is calculated relative to its corresponding reference category,

$$\begin{aligned}\delta_F &= \log(11.4/14.4), & \delta_A &= \log(4.6/12.9), \\ \delta_{35} &= \log(15.7/11.4), & \delta_{40} &= \log(10.7/11.4), \\ \delta_P &= \log(9.1/14.7).\end{aligned}$$

These coefficients represent differences in treatment log odds ratios between subgroup categories and are not scaled by 0.6. Define

$$h(X_i) = \delta_F F_i + \delta_A A_i + \delta_{35} B_{35,i} + \delta_{40} B_{40,i} + \delta_P P_i.$$

The semaglutide model is

$$p_1(X_i) = \text{logit}^{-1}(\alpha_0 + m(X_i) + \alpha_{\text{Pub}} + h(X_i)).$$

The calibration intercept α_{Pub} is added to the placebo linear predictor $\alpha_0 + m(X_i)$ and is chosen using the same calibration population and bisection method so that the mean semaglutide response probability equals r_1 .

Continuous. Age is standardised as

$$Z_i = \frac{\text{age}_i - \mu_{\text{age}}}{\sigma_{\text{age}}},$$

where μ_{age} and σ_{age} are the mean and population standard deviation of age in the calibration population. These values remain fixed across simulated trials. The semaglutide model is

$$p_1(X_i) = \text{logit}^{-1}(\alpha_0 + m(X_i) + \alpha_C - 0.5Z_i).$$

Here, Z_i acts as a continuous treatment-effect modifier, with coefficient -0.5 decreasing the semaglutide treatment effect on the log-odds scale as age increases, i.e. older participants are assumed to experience a smaller treatment effect. The calibration intercept α_C is added to the placebo linear predictor $\alpha_0 + m(X_i)$ and is chosen using the same calibration population and bisection method so that the mean semaglutide response probability equals r_1 .

For the joint subgroup, published, and continuous scenarios, the target mean semaglutide response probability r_1 applies to the calibration population rather than being imposed on each simulated trial or the evaluation population.

4.4 Estimation methods

4.4.1 Classical logistic regression

The classical logistic regression model uses treatment, the five subgroup indicators in S_i , and their treatment interactions. Let $T_i = 0$ denote placebo and $T_i = 1$ denote semaglutide, and let $Y_i = Y_i(T_i)$ denote the observed response. The model is

$$\text{logit}(\mathbb{P}(Y_i = 1 \mid T_i, X_i)) = \beta_0 + \beta_T T_i + \boldsymbol{\beta}^\top S_i + T_i \boldsymbol{\gamma}^\top S_i.$$

The model therefore contains an intercept, a treatment coefficient, five subgroup coefficients, and five treatment-by-subgroup interaction coefficients. The reference participant is male, aged under 65 years, with BMI 30 to under 35 and normoglycaemia. Age and BMI are included as the categories defined in Section 4.2 rather than as continuous variables.

The model is fitted using `statsmodels.GLM` (Seabold and Perktold, 2010) with a binomial family and logit link, a maximum of 100 fitting iterations, and the HC0 robust covariance estimator. For each participant in the evaluation population, the fitted model predicts the response probability under both treatment assignments. The estimated CATE is

$$\hat{\tau}_i = \text{logit}^{-1}(\hat{\beta}_0 + \hat{\beta}_T + (\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\gamma}})^\top S_i) - \text{logit}^{-1}(\hat{\beta}_0 + \hat{\boldsymbol{\beta}}^\top S_i).$$

For an evaluation subgroup G_g , the estimated subgroup treatment effect is the mean CATE among participants in that subgroup. The subgroup difference is

$$\hat{D} = \frac{1}{|G_1|} \sum_{i \in G_1} \hat{\tau}_i - \frac{1}{|G_0|} \sum_{i \in G_0} \hat{\tau}_i.$$

The scenario-specific evaluation subgroups are defined in Chapter 5.

Uncertainty in $\widehat{D}$ is estimated using the delta method based on the HC0 robust covariance matrix of the fitted coefficients. The 95% confidence interval is calculated as

$$\widehat{D} \pm 1.96 \widehat{\text{SE}}(\widehat{D}).$$

4.4.2 Bayesian hierarchical model (BHM)

The BHM divides participants into 24 mutually exclusive strata formed by all combinations of two sex groups, two age groups, three BMI groups, and two glycaemic-status groups. Let $k = 1, \dots, 24$ index the strata and $k(i)$ denote the stratum containing participant i . For each participant,

$$Y_i \sim \text{Bernoulli}(p_i),$$

with

$$\text{logit}(p_i) = \tilde{\beta}_{0,k(i)} \mathbb{I}(T_i = 0) + \tilde{\beta}_{1,k(i)} \mathbb{I}(T_i = 1).$$

Thus, $\tilde{\beta}_{0,k}$ and $\tilde{\beta}_{1,k}$ are the placebo and semaglutide log odds for stratum k . They follow the arm-specific hierarchical distributions

$$\tilde{\beta}_{0,k} \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad \tilde{\beta}_{1,k} \sim \mathcal{N}(\mu_1, \sigma_1^2).$$

For arm $a \in \{0, 1\}$, μ_a is the arm-specific global mean log odds and σ_a is the arm-specific between-stratum standard deviation on the log-odds scale. The primary priors are

$$\mu_0, \mu_1 \sim \mathcal{N}(0, 2^2), \quad \sigma_0, \sigma_1 \sim \text{HalfNormal}(1).$$

These priors are independent. The prior standard deviation of 2 controls uncertainty in the arm-specific global mean log odds, and the HalfNormal scale parameter of 1 controls the amount of between-stratum variation.

This structure partially pools the stratum-specific response probabilities within each treatment arm. The model adapts the arm-specific parameterisation for binary outcomes described by Wang et al. (2024), where response probabilities are modelled hierarchically within each treatment arm before treatment effects are calculated. The stratum-specific CATE is

$$\text{CATE}_k = \text{logit}^{-1}(\tilde{\beta}_{1,k}) - \text{logit}^{-1}(\tilde{\beta}_{0,k}).$$

For fitting, participant counts and outcome counts are aggregated within each treatment arm and stratum, giving an equivalent binomial likelihood.

All 24 strata are retained in each arm, including strata with no observed participants.

Posterior sampling uses PyMC (Abril-Pla et al., 2023) with two chains, 500 warm-up iterations, and 500 retained draws per chain, with a target acceptance probability of 0.95. The convergence criteria are met when the maximum $\hat{R} < 1.05$, the minimum bulk effective sample size is at least 100, and no divergent transitions occur.

At each posterior draw b , evaluation participant i is assigned the CATE of their stratum,

$$\tau_i^{(b)} = \text{CATE}_{k(i)}^{(b)}.$$

The posterior mean across retained draws gives $\hat{\tau}_i$. Subgroup treatment effects and subgroup differences are calculated within each posterior draw using the evaluation population. The posterior mean gives $\hat{D}$, and the 2.5th and 97.5th percentiles give the equal-tailed 95% credible interval.

Prior sensitivity is assessed by varying one prior component at a time. The prior standard deviation for μ_a takes values 1, 2, and 5 while the HalfNormal scale for σ_a remains fixed at 1. The HalfNormal scale for σ_a takes values 0.5, 1, and 2 while the prior standard deviation for μ_a remains fixed at 2. This gives five distinct prior specifications, including the primary specification, with 10 replications per scenario. The sensitivity results are presented in Chapter 5.

4.4.3 Causal forest

The causal forest uses EconML’s `CausalForestDML` (Battocchi et al., 2019) with continuous age, sex, continuous BMI, and prediabetes as predictors. Treatment and outcome are specified as discrete. The method first estimates the expected outcome and the probability of receiving treatment given baseline covariates using nuisance models. Three-fold cross-fitting splits the data into three folds. For each fold, the nuisance models are fitted on the other two folds and used to generate predictions for the remaining fold.

The outcome and treatment nuisance models are implemented using scikit-learn (Pedregosa et al., 2011). The outcome nuisance model uses a random forest with 100 trees, a minimum leaf size of 10, and two randomly selected predictors considered at each split. The treatment nuisance model uses a `DummyClassifier` with `strategy="prior"`, which estimates the treatment probability as the proportion of treated participants in the two training

folds and assigns this probability to all participants in the remaining fold.

The final causal forest contains 400 trees, each with a minimum leaf size of 20 and a maximum depth of 5. Each tree is fitted to a random 45% subsample of the trial data using the `mse` splitting criterion. Honest fitting separates observations used to determine splits from those used to estimate treatment effects within leaves, following Wager and Athey (2018).

The final hyperparameter specification is selected from 12 combinations of minimum leaf sizes of 5, 10, and 20, subsample proportions of 45% and 50%, and either unrestricted depth or a maximum depth of 5. Each specification is evaluated using 30 replications per scenario. The selected specification minimises the equally weighted mean CATE RMSE across the four scenarios. Tuning uses trial datasets generated separately from those in the main simulation. Full tuning results are reported in Appendix A.1.

For each participant in the evaluation population, the fitted causal forest directly estimates $\hat{\tau}_i$. Subgroup treatment effects are calculated by averaging these CATE estimates within each subgroup, using the same definition of $\hat{D}$ as the classical logistic regression model.

Uncertainty in the subgroup difference is estimated using the subgroup standard errors returned by the causal forest model. Let s_g denote the standard error for subgroup g . Assuming zero covariance between the two subgroup estimates,

$$\widehat{\text{SE}}(\hat{D}) = \sqrt{s_1^2 + s_0^2}.$$

Thus, the 95% confidence interval is

$$\hat{D} \pm 1.96 \widehat{\text{SE}}(\hat{D}).$$

4.5 Evaluation metrics

CATE root mean squared error (RMSE) evaluates individual CATE estimation accuracy for the three methods in the evaluation population. For each usable replication, the squared differences between the estimated and true CATEs are averaged across the 1,000 evaluation participants. These mean squared errors are then averaged across usable replications before taking the square root. Similarly, subgroup-difference RMSE is the square root of the mean squared difference between the estimated and true subgroup differences across usable replications.

Empirical 95% interval coverage is the percentage of usable Monte Carlo replications in which the corresponding 95% interval contains the true subgroup difference. The classical logistic regression model and causal forest use confidence intervals, and the BHM uses a 95% credible interval. Both RMSE quantities are on the risk-difference scale. Lower RMSE indicates greater estimation accuracy, and interval coverage is better when it is closer to the nominal 95% level.

The CATE and subgroup-difference estimates from the three methods are evaluated across the four scenarios in Chapter 5.

Chapter 5

RESULTS

5.1 True CATE patterns

Figure 5.1 shows the true response probabilities and CATEs in the fixed evaluation population. The dashed line labelled Population ATE denotes the common calibration target Δ . Because calibration is performed using the calibration population, the mean CATE in the evaluation population does not necessarily equal Δ in every scenario. In the no-HTE scenario, every participant has the same CATE, Δ , despite having different response probabilities in the two arms. In contrast, the joint subgroup, published, and continuous scenarios show variation in CATEs across participants.

The scenario-specific subgroup difference compares mean CATEs using the definition in Chapter 4. A positive value means that G_1 has a larger mean CATE than G_0 , whilst a negative value means that it has a smaller mean CATE. For the no-HTE and published scenarios, G_1 contains participants aged 65 or older and G_0 contains those aged under 65. The true subgroup differences are zero and -0.1766 , respectively. Thus, there is no subgroup difference under no HTE, whereas participants aged 65 or older have a smaller mean CATE in the published scenario. For the joint subgroup scenario, participants with both BMI of at least 40 and prediabetes are compared to all others. The true subgroup difference is 0.0943 , i.e. this subgroup has a larger mean CATE. For the continuous scenario, the upper age quartile is compared to the lower age quartile. The true subgroup difference is -0.1498 , i.e. participants in the upper age quartile have a smaller mean CATE.

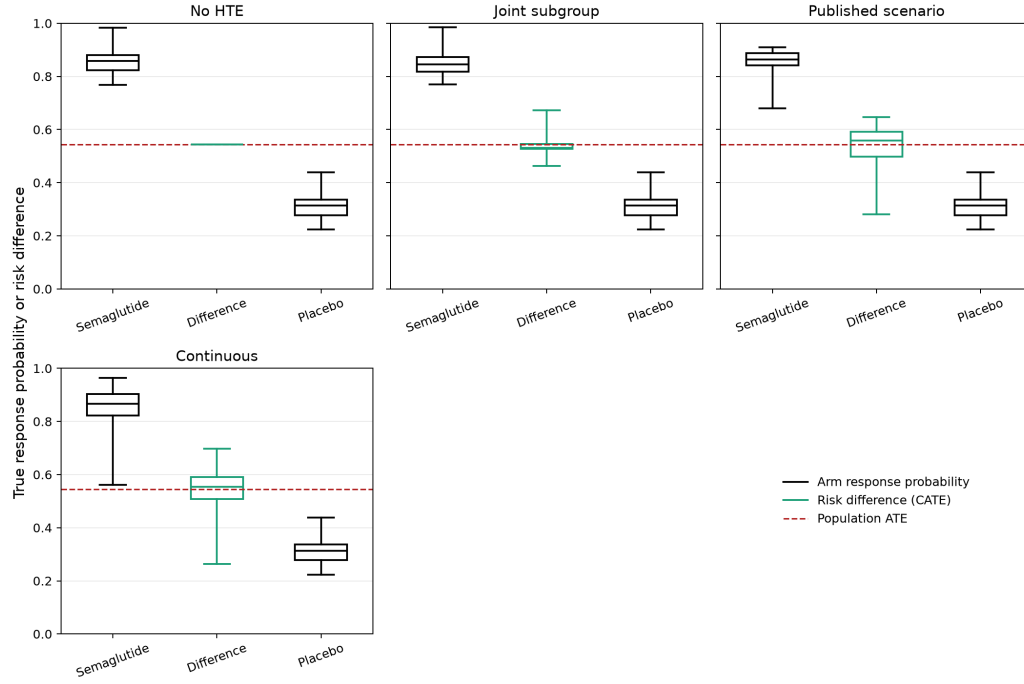


Figure 5.1: True semaglutide and placebo response probabilities and CATEs across the four scenarios in the evaluation population. Boxes show the interquartile range, central lines show medians, and whiskers show minima and maxima.

5.2 Main simulation performance

All 500 replications per scenario are usable for the classical logistic regression model and causal forest. For the BHM, the numbers of usable replications are 493, 496, 494, and 497 in the no-HTE, joint subgroup, published, and continuous scenarios, respectively. The remaining BHM fits do not meet the convergence criteria in Chapter 4 and are excluded from all three performance summaries.

Figure 5.2 compares CATE RMSE, subgroup-difference RMSE, and empirical 95% interval coverage in the four scenarios. Overall, performance depends on both the scenario and the evaluation metric. The BHM has the lowest CATE RMSE in the first three scenarios, but the causal forest model performs best in the continuous scenario. For subgroup-difference estimation, the best-performing method changes between scenarios, as discussed

below.

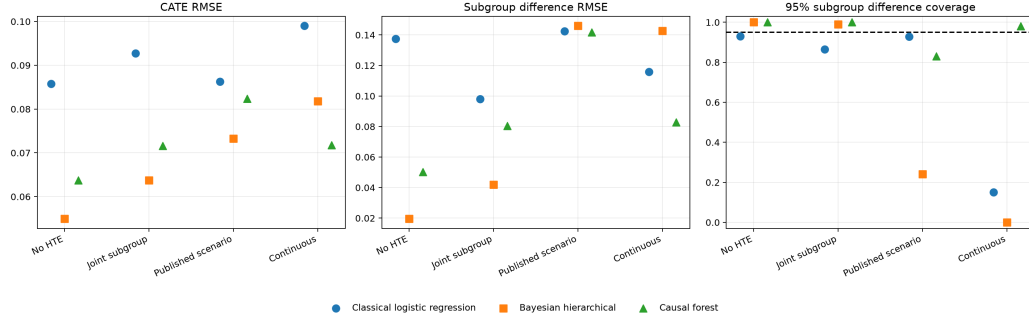


Figure 5.2: Main simulation performance in the four scenarios. The dashed line marks nominal 95% interval coverage.

5.2.1 Individual CATE estimation

The BHM has the lowest CATE RMSE in the no-HTE, joint subgroup, and published scenarios (Table 5.1). Causal forest has the next lowest RMSE in each of these scenarios, whilst the classical logistic regression model has the highest. In the no-HTE scenario, CATE RMSE is 0.0550 for the BHM, compared to 0.0637 for causal forest and 0.0858 for the classical logistic regression model. The difference between the BHM and causal forest is slightly smaller in the joint subgroup scenario.

In the continuous scenario, causal forest has the lowest CATE RMSE at 0.0718, followed by the BHM at 0.0818 and the classical logistic regression model at 0.0990. The continuous scenario gives the highest CATE RMSE for both the BHM and classical logistic regression model, whereas causal forest has its highest CATE RMSE in the published scenario.

Table 5.1: Individual-level CATE RMSE			
Scenario	Classical	BHM	Causal forest
No HTE	0.0858	0.0550	0.0637
Joint subgroup	0.0927	0.0637	0.0717
Published scenario	0.0862	0.0732	0.0823
Continuous	0.0990	0.0818	0.0718

5.2.2 Subgroup-difference estimation

The BHM also has the lowest subgroup-difference RMSE in the no-HTE and joint subgroup scenarios (Table 5.2). In the no-HTE scenario, its RMSE is 0.0196, compared to 0.0502 for causal forest and 0.1375 for the classical logistic regression model. Empirical coverage is 100.0% for the BHM and causal forest, and 93.0% for the classical logistic regression model.

Table 5.2: Subgroup-difference RMSE and empirical 95% interval coverage

Scenario	Classical	BHM	Causal forest
<i>Subgroup-difference RMSE</i>			
No HTE	0.1375	0.0196	0.0502
Joint subgroup	0.0980	0.0418	0.0803
Published scenario	0.1425	0.1459	0.1417
Continuous	0.1158	0.1427	0.0828
<i>Empirical 95% interval coverage (%)</i>			
No HTE	93.0	100.0	100.0
Joint subgroup	86.4	99.0	100.0
Published scenario	92.8	24.1	83.0
Continuous	15.0	0.0	98.0

Additional subgroup-difference bias, coverage MCSEs, and interval summaries are reported in Appendix Tables A.2 and A.3.

In the joint subgroup scenario, the BHM again performs best, with the lowest subgroup-difference RMSE of 0.0418 and empirical coverage of 99.0%. Causal forest has the next lowest RMSE at 0.0803, with coverage of 100.0%, whilst the classical logistic regression model has an RMSE of 0.0980 and coverage of 86.4%. The BHM therefore estimates the subgroup difference most accurately in this scenario, whereas the classical logistic regression model has less than 95% coverage.

In the published scenario, subgroup-difference RMSE is similar across methods, ranging from 0.1417 to 0.1459, but empirical coverage differs considerably. The classical logistic regression model has coverage closest to the nominal level, at 92.8%, compared to 83.0% for causal forest and 24.1% for the BHM. The near-nominal coverage of the classical logistic regression model is consistent with its interaction structure being more closely aligned with the data-generating model in this scenario.

Causal forest performs best in the continuous scenario, with the lowest subgroup-difference RMSE of 0.0828 and coverage of 98.0%. The classical logistic regression model has an RMSE of 0.1158 and coverage of 15.0%, and the BHM has an RMSE of 0.1427 and coverage of 0.0%. The results therefore show that causal forest recovers the subgroup difference more accurately in the continuous scenario, whereas the classical logistic regression model and BHM perform poorly for this subgroup comparison.

Overall, the results show that performance depends on how well each method can represent the underlying HTE pattern. The classical logistic regression model performs relatively well when its interaction structure is aligned with the data-generating model, as in the published scenario, but performs poorly when the true heterogeneity is not captured by its pre-specified interaction terms, especially in the continuous scenario. The BHM performs well in the no-HTE and joint subgroup scenarios, and causal forest performs best when heterogeneity follows the more flexible continuous pattern. These results suggest that methods capable of adapting to the form of heterogeneity can offer advantages when the true HTE structure is not known in advance.

5.3 BHM prior sensitivity

The five prior specifications show the same broad pattern of BHM performance (Table 5.3), with the full results reported in Appendix Table A.4. Across these specifications, subgroup-difference RMSE is lower in the no-HTE and joint subgroup scenarios than in the published and continuous scenarios. Coverage is 100.0% in the no-HTE and joint subgroup scenarios, between 10.0% and 22.2% in the published scenario, and 0.0% in the continuous scenario. Therefore, changing either prior component does not remove the poor subgroup-difference coverage in the latter two scenarios. For each performance metric, the range in Table 5.3 is calculated separately across the five prior specifications, so the minimum and maximum values within a row do not necessarily come from the same specification.

Table 5.3: BHM performance across the five prior specifications

Scenario	CATE RMSE	Subgroup-difference RMSE	Coverage (%)
No HTE	0.0527 to 0.0626	0.0239 to 0.0446	100.0
Joint subgroup	0.0594 to 0.0650	0.0223 to 0.0248	100.0
Published scenario	0.0705 to 0.0787	0.1459 to 0.1539	10.0 to 22.2
Continuous	0.0793 to 0.0824	0.1444 to 0.1459	0.0

Changing the prior standard deviation for μ_a changes how strongly the prior favours arm-specific global mean log odds close to zero. Changing the HalfNormal scale for σ_a changes how much variation between strata is favoured by the prior, and can therefore affect how strongly information is pooled across strata. Although the evaluation metrics vary between prior specifications, the overall performance pattern across scenarios remains similar. Since the sensitivity analysis uses only 10 replications per scenario and specification, differences between prior specifications should not be overinterpreted.

Chapter 6

DISCUSSION & CONCLUSION

6.1 Summary

This study compares the classical logistic regression model, BHM, and causal forest for estimating HTE in synthetic STEP UP-inspired obesity trials. The comparison assesses individual-level CATEs and prespecified subgroup differences on the risk-difference scale under the no-HTE, joint subgroup, published, and continuous scenarios.

The BHM has the lowest individual-level CATE RMSE in the no-HTE, joint subgroup, and published scenarios. It also has the lowest subgroup-difference RMSE in the no-HTE and joint subgroup scenarios, although its interval coverage is above the nominal level in both. Causal forest has the lowest individual-level CATE RMSE and subgroup-difference RMSE in the continuous scenario, where its subgroup-difference coverage is close to the nominal level.

The published scenario shows that good individual CATE estimation does not necessarily imply reliable subgroup-difference inference. Subgroup-difference RMSE is similar across methods, but coverage is 92.8% for the classical logistic regression model compared to 24.1% for the BHM. The near-nominal coverage of the classical logistic regression model is consistent with its interaction structure being closely aligned with the data-generating model.

6.2 Limitations

The data are fully synthetic, and the information used to construct them is extracted from the STEP UP summary statistics. Age, sex, BMI, and pre-diabetes are generated independently, and all covariates and outcomes are complete. These assumptions simplify the trial population and do not reproduce its joint covariate distribution or possible missingness. The findings therefore describe performance under the specified simulation process and do not establish HTE in the actual STEP UP population.

The continuous scenario may favour causal forest because it has access to continuous age, whereas the classical logistic regression model and BHM use age categorised at 65 years. Its superior performance in this scenario may therefore reflect both greater modelling flexibility and its use of continuous age rather than a binary age category. Hence, the results should not be interpreted as showing that causal forest is inherently superior to classical or Bayesian models when treatment-effect heterogeneity varies continuously with age.

The BHM models CATEs through 24 strata and partially pools response probabilities within each treatment arm. Its performance therefore depends on the stratum and prior specifications that influence the degree of pooling. The prior sensitivity analysis shows the same broad performance pattern across the five specifications, including poor subgroup-difference coverage in the published and continuous scenarios. However, only 10 replications are used per scenario and specification, which may limit the precision of comparisons between prior specifications.

For causal forest, the subgroup-difference confidence intervals omit the covariance between the two subgroup estimates. This approximation may affect the reported coverage, so the high coverage should be interpreted more cautiously.

6.3 Implications and future work

The results suggest that method choice should depend on the expected form of HTE and the intended estimation target. The classical logistic regression model is useful when relevant effect modifiers and interactions can be pre-specified, but its performance deteriorates when the assumed structure does not capture the true heterogeneity. The BHM performs well when treatment

effects can be represented through predefined strata and partial pooling is beneficial. Causal forest is useful when the HTE structure is less clearly specified.

Future work can compare causal forest to classical and Bayesian models that also use continuous age, allowing a fairer comparison in the continuous scenario. Further simulations can generate correlated covariates using joint distributions, examine larger sample sizes and different treatment allocation ratios, and consider additional HTE patterns, including more complex or nonlinear forms. More replications in the prior sensitivity analysis may improve the precision of comparisons between prior specifications.

6.4 Conclusion

No single method performs best in every scenario or for every metric. Overall, the findings suggest that model performance depends on the HTE pattern, how well it is captured by the model, and whether the focus is on individual CATEs or subgroup differences.

Appendix A

ADDITIONAL RESULTS

A.1 Causal-forest tuning

Table A.1 reports the results for the 12 candidate causal-forest hyperparameter specifications. Each candidate is evaluated using 30 replications per scenario, and the selection criterion is the equally weighted mean CATE RMSE across the four scenarios.

Table A.1: Causal-forest tuning results				
Minimum leaf size	Subsample proportion	Mean CATE RMSE		
		Unrestricted depth	Maximum depth 5	
5	45%	0.1057	0.0781	
5	50%	0.1097	0.0784	
10	45%	0.0863	0.0742	
10	50%	0.0896	0.0749	
20	45%	0.0713	0.0694	
20	50%	0.0730	0.0700	

The selected specification has a minimum leaf size of 20, a subsample proportion of 45%, and a maximum depth of 5, giving the lowest mean CATE RMSE of 0.0694. This is the specification used in the main simulation.

A.2 Subgroup-difference summaries

Table A.2 reports subgroup-difference bias and Monte Carlo standard errors (MCSEs) for coverage to supplement the RMSE and coverage results in Chapter 5.

Table A.2: Subgroup-difference bias and coverage MCSE

Scenario	Classical	BHM	Causal forest
<i>Subgroup-difference bias</i>			
No HTE	−0.0050	0.0014	−0.0023
Joint subgroup	−0.0628	−0.0099	−0.0727
Published scenario	−0.0043	0.1425	0.1270
Continuous	0.1094	0.1425	0.0452
<i>Coverage MCSE (%)</i>			
No HTE	1.1	0.0	0.0
Joint subgroup	1.5	0.4	0.0
Published scenario	1.2	1.9	1.7
Continuous	1.6	0.0	0.6

Bias is on the risk-difference scale. Coverage MCSE is $100\sqrt{\hat{c}(1 - \hat{c})/R}$, where $\hat{c}$ is empirical coverage and R is the number of usable replications. Under this formula, an MCSE of zero occurs when empirical coverage is 0% or 100%.

Table A.3 compares the empirical standard deviation (SD) of subgroup-difference estimates across usable replications to the average uncertainty reported within each replication. The latter is measured by the standard error (SE) for classical logistic regression and causal forest and by the posterior SD for the BHM. The table also reports the mean 95% interval width.

Table A.3: Subgroup-difference variability and interval width

Method	Usable replications	Empirical SD	Mean SE or posterior SD	Mean 95% interval width
<i>No HTE</i>				
Classical	500	0.1375	0.1335	0.5234
BHM	493	0.0196	0.0450	0.1834
Causal forest	500	0.0502	0.0977	0.3830
<i>Joint subgroup</i>				
Classical	500	0.0752	0.0723	0.2834
BHM	496	0.0407	0.0562	0.2235
Causal forest	500	0.0342	0.0937	0.3672
<i>Published scenario</i>				
Classical	500	0.1425	0.1388	0.5443
BHM	494	0.0315	0.0501	0.2012
Causal forest	500	0.0629	0.1045	0.4098
<i>Continuous</i>				
Classical	500	0.0380	0.0343	0.1344
BHM	497	0.0079	0.0133	0.0538
Causal forest	500	0.0694	0.1004	0.3936

All quantities except replication counts are on the risk-difference scale. Mean uncertainty is the mean SE for classical logistic regression and causal forest, and the mean posterior SD for the BHM. Mean interval width is based on the corresponding 95% confidence or credible intervals.

A.3 BHM prior sensitivity

Table A.4 reports the full results for each prior specification shown in Table 5.3. Within each scenario, all specifications are evaluated on the same 10 trial datasets, generated separately from the main simulation. Fits that do not meet the convergence criteria are excluded from the summaries.

Table A.4: BHM performance by prior specification

Prior SD for μ_a	HalfNormal scale for σ_a	Usable (out of 10)	CATE RMSE	Subgroup-difference RMSE	Coverage (%)
<i>No HTE</i>					
1	1	10	0.0588	0.0360	100.0
2	1	10	0.0603	0.0374	100.0
5	1	10	0.0597	0.0387	100.0
2	0.5	10	0.0527	0.0239	100.0
2	2	10	0.0626	0.0446	100.0
<i>Joint subgroup</i>					
1	1	10	0.0636	0.0239	100.0
2	1	10	0.0630	0.0231	100.0
5	1	10	0.0631	0.0223	100.0
2	0.5	10	0.0594	0.0239	100.0
2	2	10	0.0650	0.0248	100.0
<i>Published scenario</i>					
1	1	9	0.0762	0.1459	22.2
2	1	9	0.0749	0.1476	22.2
5	1	10	0.0747	0.1490	20.0
2	0.5	10	0.0705	0.1539	10.0
2	2	9	0.0787	0.1474	22.2
<i>Continuous</i>					
1	1	10	0.0812	0.1444	0.0
2	1	10	0.0811	0.1450	0.0
5	1	10	0.0813	0.1457	0.0
2	0.5	10	0.0793	0.1459	0.0
2	2	10	0.0824	0.1452	0.0

The prior SD for μ_a and the HalfNormal scale for σ_a apply to both treatment arms. The primary specification uses a prior SD of 2 for μ_a and a HalfNormal scale of 1 for σ_a . RMSE is on the risk-difference scale, and coverage is the empirical coverage of the 95% credible interval for the subgroup difference.

Appendix B

CODE EXCERPTS

This appendix presents selected code excerpts used for synthetic data generation, model fitting, and evaluation. The excerpts show the main implementation described in Chapters 3 and 4. The complete analysis code is available in the accompanying GitHub repository.

B.1 Synthetic data generation

The following excerpts show the generation of baseline covariates, calibration to the target response probabilities, construction of the four heterogeneity scenarios, and generation of the observed trial outcome.

```
female = random_generator.binomial(1, FEMALE_PROBABILITY, n)
prediabetes = random_generator.binomial(1, PREDIABETES_PROBABILITY, n)

age = sample_truncated_normal(
    random_generator, AGE_GENERATING_MEAN, AGE_GENERATING_SD,
    18, np.inf, n
)
bmi = sample_truncated_normal(
    random_generator, BMI_GENERATING_MEAN, BMI_GENERATING_SD,
    30, np.inf, n
)
```

```
def calibrate_intercept(base_score, target_probability):
    lower = -10.0
    upper = 10.0

    for _ in range(80):
```

```

        midpoint = (lower + upper) / 2
        if inverse_logit(
            base_score + midpoint
        ).mean() < target_probability:
            lower = midpoint
        else:
            upper = midpoint

    return (lower + upper) / 2

```

```

placebo_log_odds = (
    PLACEBO_INTERCEPT
    + subgroup_score(data, "placebo_log_or")
)
placebo_probability = inverse_logit(placebo_log_odds)

if scenario == "no_hte":
    treated_probability = placebo_probability + ATE_TARGET

elif scenario == "joint_subgroup":
    treatment_log_odds = (
        placebo_log_odds
        + JOINT_TREATMENT_INTERCEPT
        + JOINT_SUBGROUP_COEFFICIENT * joint_subgroup(data)
    )
    treated_probability = inverse_logit(treatment_log_odds)

elif scenario == "published_scenario":
    treatment_log_odds = (
        placebo_log_odds
        + PUBLISHED_TREATMENT_INTERCEPT
        + subgroup_score(data, "interaction_log_or")
    )
    treated_probability = inverse_logit(treatment_log_odds)

else:
    treatment_log_odds = (
        placebo_log_odds
        + CONTINUOUS_TREATMENT_INTERCEPT
        + CONTINUOUS_AGE_COEFFICIENT * standardised_age(data)
    )
    treated_probability = inverse_logit(treatment_log_odds)

```

```

treated_n = round(TREATMENT_PROBABILITY * len(data))
treated = np.r_[
    np.ones(treated_n, dtype=int),
    np.zeros(len(data) - treated_n, dtype=int)
]
random_generator.shuffle(treated)

```

```

data["treated"] = treated

observed_probability = np.where(
    treated == 1, treated_probability, placebo_probability
)

data["weight_loss_5pct"] = random_generator.binomial(
    1, observed_probability
)

true_cate = treated_probability - placebo_probability

```

B.2 Classical logistic regression

The following excerpts show construction of the regression design matrix with treatment-by-subgroup interactions, prediction of CATEs, and fitting of the classical logistic regression model.

```

subgroup_design = data[MODEL_FACTOR_COLUMNS]

main_design = pd.DataFrame(
    {"intercept": 1.0, "treated": data["treated"]},
    index=data.index
)

interactions = subgroup_design.mul(
    data["treated"], axis=0
).add_prefix("treated_x_")

observed_design = pd.concat(
    [main_design, subgroup_design, interactions],
    axis=1
)

fitted_model = sm.GLM(
    data["weight_loss_5pct"],
    observed_design,
    family=sm.families.Binomial()
).fit(cov_type="HC0", maxiter=100)

```

```

treated_design = make_classical_design(evaluation_data, treatment=1)
placebo_design = make_classical_design(evaluation_data, treatment=0)

treated_risk = cfg.inverse_logit(treated_design @ fitted_model.params)
placebo_risk = cfg.inverse_logit(placebo_design @ fitted_model.params)

```

```
cate = treated_risk - placebo_risk
```

B.3 Bayesian hierarchical model

The following excerpts show construction of the 24 covariate-defined strata, fitting of the arm-specific Bayesian hierarchical model, and posterior CATE estimation.

```
N_STRATA = 24

bmi_group = indicators["bmi_35_to_40"] + 2 * indicators["bmi_40"]

stratum = (
    indicators["female"] * 12
    + indicators["age_65"] * 6
    + bmi_group * 2
    + indicators["prediabetes"]
)
```

```
with pm.Model() as model:
    global_log_odds = pm.Normal(
        "global_log_odds", mu=0, sigma=2, shape=2
    )

    heterogeneity = pm.HalfNormal(
        "heterogeneity", sigma=1, shape=2
    )

    stratum_z = pm.Normal(
        "stratum_z", mu=0, sigma=1, shape=(2, N_STRATA)
    )

    log_odds = (
        global_log_odds[:, None]
        + heterogeneity[:, None] * stratum_z
    )

    probability = pm.Deterministic(
        "probability", pm.math.sigmoid(log_odds)
    )

    pm.Binomial(
        "responses", n=totals, p=probability, observed=responses
    )
```

```
posterior = pm.sample(
    draws=500, tune=500, chains=2, target_accept=0.95,
    random_seed=fit_seed
)
```

```
probability_draws = np.asarray(
    posterior.posterior["probability"], dtype=float
)

cate_draws = (
    probability_draws[:, :, 1, evaluation_stratum]
    - probability_draws[:, :, 0, evaluation_stratum]
)

cate = cate_draws.mean(axis=(0, 1))
```

B.4 Causal forest

The following excerpt shows the nuisance models, cross-fitting, causal-forest specification, and CATE prediction.

```
forest = CausalForestDML(
    model_y=RandomForestClassifier(
        n_estimators=100,
        min_samples_leaf=10,
        max_features="sqrt",
        random_state=fit_seed,
        n_jobs=1
    ),
    model_t=DummyClassifier(strategy="prior"),
    discrete_outcome=True,
    discrete_treatment=True,
    cv=3,
    n_estimators=400,
    min_samples_leaf=20,
    max_samples=0.45,
    criterion="mse",
    max_depth=5,
    honest=True,
    inference=True,
    random_state=fit_seed,
    n_jobs=1
)

forest.fit(
```

```

    data["weight_loss_5pct"].to_numpy(dtype=int),
    data["treated"].to_numpy(dtype=int),
    X=get_baseline_features(data),
    inference="auto"
)

cate = np.asarray(
    forest.effect(get_baseline_features(evaluation_data)),
    dtype=float
).reshape(-1)

```

B.5 Model evaluation

The following excerpts show how estimated CATEs and subgroup differences are compared to their known true values and how the main performance metrics are calculated.

```

subgroup_difference_result = fitted["subgroup_difference_result"]

subgroup_difference_error = (
    subgroup_difference_result["estimate"]
    - truth["subgroup_difference"]
)

cate_mse = np.mean((fitted["cate"] - truth["cate"]) ** 2)

covered = (
    subgroup_difference_result["lower"]
    <= truth["subgroup_difference"]
    <= subgroup_difference_result["upper"]
)

```

```

usable = subset[subset["success"] & subset["converged"]]

cate_rmse = np.sqrt(usable["cate_mse"].mean())

subgroup_difference_rmse = np.sqrt(
    np.mean(usable["subgroup_difference_error"] ** 2)
)

coverage = usable["subgroup_difference_covered"].mean()

```

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